

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIOBASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	AIDS-2024	Date:	

Code of Conduct

I J. Pranitha(192472073)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Pranitha

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

English Word Classes, Tagsets for English, Part of Speech Tagging

Morphological analysis, and affixes, inflection, normalization

English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Mor and bas
Counting Words, English Word Classes, Part of Speech Tagging	Mor gov moc

Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphologic and reactivation layers, word-
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional r and creating; offense/aspe

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Aim

To design a morphological processing approach that decomposes each word into its root and affixes, distinguishes between inflectional and derivational forms, and normalizes them into a common representation for efficient search engine indexing.

Step 1: Tokenization

The input words are:

- analyzing •
- analysis
- analytical
-

Step 2: Morphological Decomposition

Input Word	Root	Prefix	Suffix	Word Formation
analyzing	analyze	—	-ing	Root + Inflectional Suffix
analysis	analyze	—	-sis	Root + Derivational Suffix
analytical	analyze	—	-ical	Root + Derivational Suffix

Step 3: Classification

- Analyzing o Root: analyze o Suffix: -ing o Type: Inflectional o Meaning: Indicates an ongoing action without changing the core meaning.
- Analysis o Root: analyze o Suffix: -sis o Type: Derivational o Meaning: Changes the verb into a noun representing the process or result.
- Analytical o Root: analyze o Suffix: -ical o Type: Derivational o Meaning: Changes the noun into an adjective describing characteristics.

Step 4: Normalization

All forms are normalized to the common base form:

Normalized Root = analyze

Morphological Structure

Word	Morphological Structure	Normalized Output
analyzing	analyze + ing	analyze
analysis	analyze + sis	analyze
analytical	analyze + ical	analyze

Conclusion

The search engine decomposes each word into its root and affixes. Inflectional forms preserve grammatical meaning, whereas derivational forms create new lexical categories. By normalizing all variants to the root "analyse", retrieval accuracy improves because related words are indexed under one representation.

2. An AI-based sentiment analysis platform receives product reviews containing words like "disagree", "agreement", and "agreeable". The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

Aim

To design a morphological parsing strategy that analyzes the words "disagree", "agreement", and "agreeable" by identifying their prefixes, suffixes, and root forms. The strategy should distinguish derivational changes and normalize the words to improve semantic understanding in an AI-based sentiment analysis system.

Introduction

In sentiment analysis, users often express opinions using different forms of the same word. For example, the words agree, disagree, agreement, and agreeable are morphologically related but convey different meanings and sentiments. A morphological parser breaks words into meaningful units (morphemes), identifies the root word, and determines how prefixes and suffixes modify the original meaning. By converting different word forms into a common representation while preserving semantic changes, the AI system can classify sentiments more accurately.

Morphological Parsing Strategy

The following steps are performed by the parser: **Step**

1: Tokenization

The review text is divided into individual words.

Example Input

disagree agreement agreeable

↓

["disagree", "agreement", "agreeable"]

Step 2: Morpheme Identification

Each word is divided into: •

Prefix

- Root (Base Form)
- Suffix

The parser checks whether the word contains any known prefixes or suffixes.

Step 3: Morphological Parsing

Word 1: disagree

Parsing dis +

agree

- Prefix = dis-
- Root = agree
- Suffix = None

Meaning

The prefix **dis-** reverses the meaning of the root.

Transformation Type

Derivational Morphology Reason:

The prefix changes the meaning from positive agreement to disagreement.

Normalized Root agree

Semantic Interpretation

Negative opinion

Sentiment

Negative

Word 2: agreement

Parsing agree +

ment

- Prefix = None
- Root = agree
- Suffix = -ment

Meaning

The suffix **-ment** converts the verb into a noun.

Transformation Type

Derivational Morphology Reason:

The grammatical category changes from verb to noun.

Normalized Root agree

Semantic Interpretation

Represents the state or act of agreeing.

Sentiment

Generally Positive / Neutral

Word 3: agreeable

Parsing agree

+ able

- Prefix = None • Root = agree
- Suffix = -able

Meaning

The suffix **-able** changes the word into an adjective meaning "pleasant" or "willing to agree."

Transformation Type

Derivational Morphology Reason:

The suffix changes the grammatical category from verb to adjective. Normalized

Root

agree

Semantic Interpretation

Positive personality or pleasant nature.

Sentiment

Positive

Morphological Analysis Table

Input Word	Prefix	Root	Suffix	Transformation	Morphology Type	Semantic Meaning	Sentiment
disagree	dis-	agree	—	Prefix changes meaning	Derivational	Opposite of agree	Negative
agreement	—	agree	-ment	Verb → Noun	Derivational	State of agreeing	Positive/Neutral
agreeable	—	agree	-able	Verb → Adjective	Derivational	Pleasant or cooperative	Positive

Normalized Output

Original Word	Normalized Root
disagree	agree
agreement	agree
agreeable	agree

The sentiment analysis system stores all three words under the common root "**agree**", while also preserving the effect of prefixes and suffixes to distinguish their meanings.

Morphological Structure

disagree

|
| — Prefix : dis- —
Root : agree agreement

|
| — Root : agree —
Suffix : -ment

agreeable

|
| — Root : agree
| — Suffix : -able

Algorithm

1. Read each input word.
2. Check for known prefixes (e.g., dis-, un-, re-).
3. Check for known suffixes (e.g., -ment, -able, -ing, -ed).
4. Separate the word into prefix, root, and suffix.
5. Identify whether the change is inflectional or derivational.
6. Normalize the word to its root form.
7. Preserve semantic changes introduced by prefixes and suffixes.
8. Use the normalized root and semantic information for sentiment classification.

Final Parsed Output

Word	Morphological Structure	Root	Type	Meaning	Sentiment
disagree	dis + agree	agree	Derivational	Opposite of agree	Negative
agreement	agree + ment	agree	Derivational	Act or state of agreeing	Positive/Neutral
agreeable	agree + able	agree	Derivational	Pleasant or cooperative	Positive

Conclusion

The morphological parsing strategy successfully decomposes disagree, agreement, and agreeable into their constituent morphemes by identifying prefixes, suffixes, and the common root agree. All three words undergo derivational morphological transformations, as they either alter the meaning or change the grammatical category of the base word. By normalizing each word to the root while preserving the semantic impact of derivational affixes, the sentiment analysis platform can accurately distinguish negative, neutral, and positive expressions, resulting in more consistent and reliable sentiment interpretation across different word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

Aim

To design a morphology-based normalization mechanism that analyzes the words "govern", "government", and "governance" by identifying their root and affixes, distinguishing derivational changes, and mapping them to a common base form for efficient topic modeling and document clustering.

Introduction

In text analytics and news aggregation systems, the same concept may appear in different morphological forms. For example, govern, government, and governance all originate from the same root but differ in grammatical category and meaning. If these words are treated as separate terms, the system may reduce the accuracy of topic modeling and document clustering.

A morphology-based normalization mechanism decomposes each word into its root and affixes, identifies derivational transformations, and converts related words into a common base representation while preserving semantic information.

Morphology-Based Normalization Mechanism

The normalization process consists of the following stages:

Step 1: Tokenization

The document is divided into individual words.

Example Input govern

government governance

↓

["govern", "government", "governance"]

Step 2: Morpheme Identification Each

word is analyzed to identify:

- Root (Base Form)
- Prefix (if present)
- Suffix (if present)

The parser compares each word against a morphological dictionary and known affix rules.

Step 3: Morphological Decomposition

Word 1: govern

Parsing govern

- Prefix = None
- Root = **govern**
- Suffix = None

Meaning

The word itself is the base verb meaning to rule or administer.

Transformation Type

Base Word (No Derivational Change)

Normalized Form

govern

Word 2: government

Parsing govern +
ment

- Prefix = None
- Root = **govern**
- Suffix = **-ment**

Meaning

The suffix **-ment** converts the verb into a noun representing the governing body or system.

Transformation Type

Derivational Morphology Reason:

The suffix changes the grammatical category from **verb** → **noun**.

Normalized Form

govern

Word 3: governance

Parsing govern

+ ance

- Prefix = None
- Root = govern
- Suffix = -ance

Meaning

The suffix **-ance** converts the verb into a noun referring to the process or practice of governing.

Transformation Type

Derivational Morphology Reason:

The suffix changes the grammatical category from **verb** → **abstract noun**.

Normalized Form

govern

Morphological Analysis Table

Input Word	Pref i x	Root	Suffi x	Transformation	Morphol ogyType	Meaning
govern	—	gov er n	—	Base form	Base Word	To rule or administer
governm ent	—	gov er n	-men t	Verb → Noun	Derivational	Governing body orsystem
governance	—	gov er	-ance	Verb → Abstract Noun	Derivational	Process or practice of governing

		n				
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Derivational Layers govern
govern

No derivational layer.

government

govern
↓
+ -ment
↓
government

Layer 1:
Verb → Noun

governance
govern
↓
+ -ance
↓
governance

Layer 1:
Verb → Abstract Noun

Normalized Representation

Original Word	Normalized Base Form
govern	govern
government	govern
governance	govern

All three words are mapped to the common base form "govern", allowing the system to group related

documents under the same conceptual topic.

Morphological Breakdown

govern govern

└─ Root : govern

government government

└─ Root : govern
└─ Suffix : -ment

governance governance

└─ Root : govern
└─ Suffix : -ance

Normalization Algorithm

1. Read each input word.
2. Detect any prefixes and suffixes using morphological rules.
3. Separate the word into root and affixes.
4. Identify whether the transformation is base, inflectional, or derivational.
5. Remove derivational suffixes to recover the root.
6. Store the normalized root as the indexing term.
7. Preserve derivational information for semantic analysis.
8. Use the normalized representation for topic modeling and clustering.

Final Normalized Output

Word	Morphological Structure	Root	Morphology Type	Normalized Form
govern	govern	govern	Base Word	govern
government	govern + ment	govern	Derivational	govern

governance	govern + ance	govern	Derivational	govern
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Conclusion

The proposed morphology-based normalization mechanism successfully analyzes govern, government, and governance by decomposing each word into its root and suffixes, identifying derivational transformations, and mapping them to the common base form "govern". While govern is the original verb, government and governance are derived nouns formed through the suffixes -ment and -ance, respectively. By normalizing all related forms to the same root while retaining derivational information, the news aggregation system can improve topic modeling, document clustering, and retrieval accuracy by recognizing these words as semantically related concepts.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class. Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

Aim

To design a morphological parsing strategy that analyzes the words "activate", "activation", and "reactivation" by identifying prefixes, suffixes, and root forms. The strategy distinguishes derivational layers, explains how affixes affect meaning and grammatical category, and maps all words to a unified base representation for document classification and indexing.

Introduction

In document classification systems, the same concept may appear in different word forms. For example, activate, activation, and reactivation share the same root but differ in meaning and grammatical category due to the addition of prefixes and suffixes. Treating these forms as separate words can reduce the accuracy of semantic grouping and indexing.

A morphological parser separates words into prefixes, roots, and suffixes, identifies derivational transformations, and normalizes related words to a common base form while preserving their semantic relationships.

Morphological Parsing Strategy The parser follows these steps:

Step 1: Tokenization

Split the input text into individual words. Example

Input

activate activation reactivation

↓

["activate", "activation", "reactivation"]

Step 2: Morpheme Identification For each word, identify:

- Prefix (if present)
- Root (Base Form)
- Suffix (if present)

The parser compares each word with a morphological dictionary and predefined affix rules.

Step 3: Morphological Decomposition

Word 1: activate

Parsing

active + ate

- Prefix = None
- Root = active
- Suffix = -ate

Meaning

The suffix **-ate** forms the verb **activate**, meaning to make something

active. Transformation Type

Derivational Morphology Reason:

The suffix changes the word class from adjective (**active**) to verb (**activate**).

Normalized Root **active**

Word 2: activation Parsing

active + ate + ion

- Prefix = None
- Root = **active**
- Suffixes = **-ate**, **-ion**

Meaning

The suffix **-ion** converts the verb **activate** into the noun **activation**, meaning the process of making something active.

Transformation Type **Derivational**

Morphology Reason:

The grammatical category changes from verb to noun.

Normalized Root **active**

Word 3: reactivation

Parsing re + active +
ate + ion

- Prefix = **re-** • Root = **active**
- Suffixes = **-ate**, **-ion**

Meaning

- Prefix **re-** means **again**.
- Suffix **-ion** converts the verb into a noun.

Therefore, reactivation means the process of activating again.

Transformation Type **Derivational**

Morphology Reason:

The prefix changes the meaning, while the suffix changes the grammatical category.

Normalized Root **active**

Morphological Analysis Table

Input Word	Prefix	Root	Suffix(es)	Transformation	Morphology Type	Meaning
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activate	—	active	-ate	Adjective → Verb	Derivational	Make active
activation	—	active	-ate, -ion	Verb → Noun	Derivational	Process of activating
reactivation	re- -	active	-ate, -ion	Prefix changes meaning; Verb → Noun	Derivational	Process of activating again

Effect of Prefix and Suffix

Prefix (re-)

- Adds the meaning **again** or **once more**.
- Changes the semantic meaning without changing the root.

Example:

activate → reactivation Meaning
changes from:

Make active

↓

Make active again

Suffix (-ate)

- Converts **active (adjective)** into **activate (verb)**.

Example:

active → activate Word

Class:

Adjective → Verb

Suffix (-ion)

- Converts **activate (verb)** into **activation (noun)**.

Example:

activate → activation Word

Class:

Verb → Noun

Derivational Layers activate

active

↓

+ -ate

↓ activate

Layer 1:

Adjective → Verb

activation

active

↓

+ -ate

↓ activate

↓

+ -ion

↓

activation Layer

1: Adjective →

Verb Layer 2:

Verb → Noun

reactivation

active

↓

+ -ate

↓ activate

↓

+ re-

↓ reactivate

↓

+ -ion

↓

reactivation Layer 1:

Adjective → Verb Layer 2:

Prefix adds meaning ("again")

Layer 3:

Verb → Noun

Morphological Structure activate

activate

|

|— Root : active

|— Suffix : -ate

activation

activation

|

|— Root : active

|— Suffix : -ate

|— Suffix : -ion reactivation

reactivation

|

|— Prefix : re- |—

Root : active

|— Suffix : -ate

└─ Suffix : -ion

Unified Base Representation

Original Word	Normalized Root
activate	active
activation	active
reactivation	active

All three words are normalized to the common base form "**active**", enabling the document classification system to recognize them as semantically related while preserving the effects of prefixes and suffixes.

Normalization Algorithm

1. Read each input word.
2. Detect prefixes such as re-, un-, dis-, etc.
3. Detect derivational suffixes such as -ate, -ion, -ment, and -ance.
4. Separate the word into prefix, root, and suffixes.
5. Identify derivational layers and changes in word class.
6. Remove derivational affixes to obtain the normalized root.
7. Store the normalized root for indexing and semantic grouping.
8. Preserve derivational information for accurate interpretation during classification.

Final Parsed Output

Word	Morphological Structure	Root	Morphology Type	Normalized Form
activate	active + ate	active	Derivational	active
activation	active + ate + ion	active	Derivational	active
reactivation	re + active + ate + ion	active	Derivational	active

Conclusion

The proposed morphological parsing strategy effectively decomposes activate, activation, and reactivation into their prefixes, root, and suffixes while identifying each derivational layer. The suffix -ate converts the adjective active into the verb activate, the suffix -ion transforms the verb into the noun activation, and the prefix re- adds the meaning "again" to form reactivation. By normalizing all three

words to the common base form "active" and preserving the semantic effects of affixes, the document classification system can improve semantic grouping, indexing, topic identification, and retrieval accuracy.

5. A search optimization module processes input words such as "create", "creates", and "creating" to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning. Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

Aim

To design a morphology-based normalization approach that analyzes the words "create", "creates", and "creating" by identifying suffix patterns, extracting the root form, classifying inflectional transformations, and mapping all variations to a common base form for consistent search retrieval.

Introduction

In search optimization systems, users may search using different grammatical forms of the same word. For example, **create**, **creates**, and **creating** represent different forms of the same verb but carry the same core meaning. If treated as separate terms, the search engine may return incomplete or inconsistent results.

A morphology-based normalization approach identifies inflectional suffixes, removes them to recover the base form, and stores the normalized word for indexing and retrieval. Since inflectional changes only modify grammatical information (such as tense or aspect) and do not change the core meaning, all forms can be mapped to the same root.

Morphology-Based Normalization Approach

The normalization process consists of the following stages:

Step 1: Tokenization

Split the input into individual words. Example

Input

create creates creating

↓

["create", "creates", "creating"]

Step 2: Morpheme Identification For

each word, identify:

- Prefix (if present)
- Root (Base Form)

- Inflectional Suffix

The parser checks for common inflectional endings such as:

- -s
- -es
- -ing
- -ed

Step 3: Morphological Decomposition

Word 1: create Parsing

create

- Prefix = None
- Root = **create**
- Suffix = None

Meaning

This is the original base verb.

Transformation Type

Base Form Normalized

Root create

Word 2: creates Parsing

create + s

- Prefix = None
- Root = **create**
- Suffix = -s

Meaning

The suffix -s indicates the third-person singular present tense.

Example:

He **creates** designs.

Transformation Type Inflectional

Morphology Reason:

The suffix changes only the grammatical form and does not change the meaning or word class.

Normalized Root create

Word 3: creating Parsing

create + ing

- Prefix = None
- Root = **create**
- Suffix = -ing

Meaning

The suffix -ing represents the present participle/progressive aspect.

Example:

She is **creating** a document.

Transformation Type Inflectional

Morphology Reason:

The suffix expresses an ongoing action without changing the meaning or grammatical category.

Normalized Root create

Morphological Analysis Table

Input Word	Prefix	Root	Suffix	Transformation	Morphology Type	Meaning
create		create		Base Form Normalized		
creates		create	-s	Inflectional		
creating		create	-ing	Inflectional		

create	—	crea te	—	Base Form	Base Word	To produce or makesomething
creates	—	crea te	-s	Present Tense (3rd PersonSingular)	Inflectional	Same meaning

creating	—	crea te	-ing	Present Participle / Progressive Aspect	Inflectional	Same meaning
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Suffix Pattern Analysis

Suffix -s

- Indicates third-person singular present tense.
- Does not change the meaning.
- Does not change the word class.

Example: create

→ creates

Verb → Verb

Suffix -ing

- Indicates continuous or progressive aspect.
- Does not change the meaning.
- Does not change the word class.

Example: create

→ creating

Verb → Verb

Morphological Breakdown

create

create

└─ Root : create

creates

creates

└─ Root : create
└─ Suffix : -s

creating

creating

└─ Root : create
└─ Suffix : -ing

Transformation Classification\

Word	Transformation	Type
create	Base Form	Base Word
creates	Third-person singular present tense	Inflectional
creating	Present participle / Progressive aspect	Inflectional

Normalized Representation

Original Word	Normalized Root
create	create
creates	create
creating	create

All three words are normalized to the common base form "create", ensuring that search queries retrieve all relevant documents regardless of tense or aspect.

Normalization Algorithm

1. Read each input word.
2. Identify inflectional suffixes such as -s, -es, -ing, and -ed.
3. Separate the suffix from the root.
4. Verify that the suffix represents an inflectional change.
5. Remove the inflectional suffix to recover the base form.
6. Store the normalized root for indexing.
7. Preserve grammatical information if needed for language processing.
8. Use the normalized root for search optimization and document retrieval.

Final Parsed Output

Word	Morphological Structure	Root	Morphology Type	Normalized Output
create	create	creat e	Base Form	create
creates	create + s	creat e	Inflectional	create
creatin g	create + ing	creat e	Inflectional	create

Conclusion

The proposed morphology-based normalization approach effectively processes create, creates, and creating by identifying their suffix patterns, extracting the common root, and classifying the transformations as inflectional morphology. The suffix -s marks the third-person singular present tense, while -ing denotes the present participle or progressive aspect. Since these transformations only affect grammatical form and do

not alter the core meaning or word class, all three words are normalized to the base form "create". This normalization improves search optimization, indexing, and information retrieval by ensuring that all grammatical variations are treated as the same semantic concept.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. /Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub . Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____